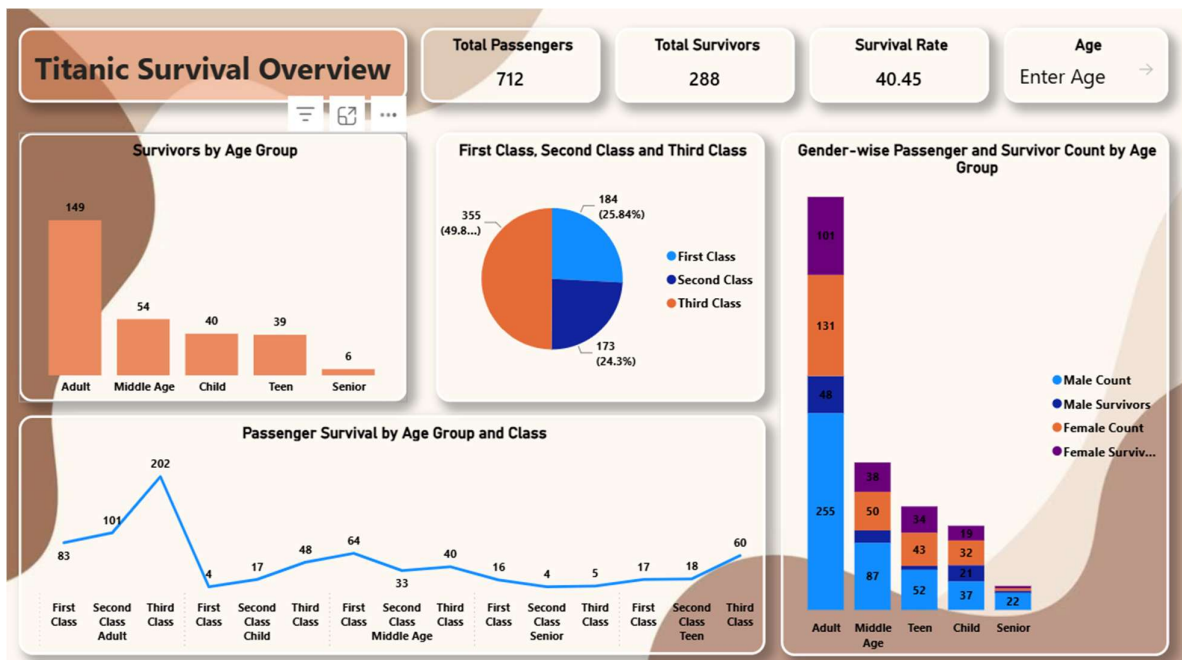


Report

A report is a comprehensive document that provides a detailed and structured account of data analysis, findings, and insights. It is typically used for in-depth analysis, documentation, and communication of results. Reports are suitable for a diverse audience, including decision-makers, analysts, and stakeholders who need a comprehensive understanding of the data.

Designing a report in Power BI involves connecting to data sources, creating visualizations like charts and graphs, customizing their appearance and interactivity, organizing them logically on the canvas, formatting elements for consistency and clarity, and optionally creating dashboards for a summarized view. Throughout the process, it's essential to consider the audience's needs and ensure the report effectively communicates insights from the data. Finally, iterate based on feedback to continually improve the report's design and usefulness.



Dashboard - 1

Titanic Survival Overview

Overall Dashboard Description:

The Titanic Survival Overview dashboard provides a comprehensive analysis of passenger survival patterns using the Titanic dataset. It highlights key survival statistics across different age groups, passenger classes, and genders, helping users understand the major factors that influenced survival during the disaster. The dashboard includes interactive KPI cards, charts, and filters for deeper exploration of passenger demographics and survival trends.

Dashboard Components

- Total Passengers: Displays the total number of passengers (712).
- Total Survivors: Shows the total number of survivors (288).
- Survival Rate: Indicates the overall survival percentage (40.45%).
- Age Filter: Allows users to filter the entire dashboard based on passenger age.

Key Visualizations

- Survivors by Age Group: Compares the number of survivors across Adult, Middle Age, Child, Teen, and Senior categories.
- Passenger Class Distribution: Shows survivor distribution among First, Second, and Third Class passengers.
- Gender-wise Passenger & Survivor Analysis: Compares male and female passenger counts with their corresponding survivor counts across age groups.
- Passenger Survival by Age Group and Class: Displays how survival varies across different age groups within each passenger class.

Key Insights

- Adults recorded the highest number of survivors (149).
- Children achieved the highest survival rate, indicating evacuation priority.
- Female passengers survived at a much higher rate than male passengers.
- Third Class contained the largest number of surviving passengers due to its larger passenger population.
- Passenger age, gender, and class were the primary factors affecting survival.

Conclusion

The Titanic Survival Overview dashboard enables users to quickly understand survival trends through interactive visualizations and KPIs. It demonstrates that gender, passenger class, and age group significantly influenced survival outcomes, making the dashboard an effective tool for exploratory data analysis and decision-making.

Dashboard – 2



Dashboard – 2

Titanic Fatality Analysis

Overall Dashboard Description

The Titanic Fatality Analysis dashboard provides a detailed examination of passenger deaths during the Titanic disaster. It focuses on fatality trends based on gender, age group, passenger class, family category, and embarkation port. The dashboard helps identify which passenger groups were most affected and the key factors contributing to the high death rate.

Dashboard KPIs:

KPI	Value
Death Rate	59.55%
Total Deaths	424
Male Deaths	360

Overview

- Out of 712 passengers, 424 passengers lost their lives, resulting in a 59.55% death rate.
- Male passengers accounted for the majority of fatalities, with 360 deaths, while female deaths were significantly lower at 64, reflecting evacuation priorities.

Dashboard Visualizations

1. Female Deaths by Age Group

This bar chart displays female fatalities across different age groups.

Age Group	Deaths
Adult	30
Child	13
Middle Age	12
Teen	9

Insights

- Adult women recorded the highest number of female deaths.
- Children experienced fewer female fatalities, indicating that young girls had relatively better chances of survival.
- Teen and middle-aged female deaths remained comparatively low.

2. Male Deaths by Age Group

This visual compares male fatalities across age groups.

Age Group	Deaths
Adult	207
Middle Age	71
Teen	47
Senior	19

Insights

- Adult males were the most affected, accounting for nearly half of all fatalities.
- Middle-aged and teenage males also experienced high mortality.
- Male children had the lowest death count, reflecting evacuation efforts focused on children.

3. Deaths by Passenger Class

The pie chart illustrates the distribution of deaths across passenger classes.

Passenger Class	Deaths
Third Class	270

Second Class	90
First Class	64

Insights

- **Third Class passengers represented approximately 64% of all deaths**, making it the most heavily impacted class.
- First Class passengers experienced the fewest fatalities due to better access to lifeboats and higher priority during evacuation.
- Passenger class played a major role in determining survival chances.

4. Child Deaths by Family Category

This funnel chart analyzes child fatalities based on family size.

Family Category	Child Deaths
Large Family	29
Alone	13
Small Family	10

Insights

- Children traveling in **large families experienced the highest number of deaths**, likely due to evacuation difficulties.
- Children traveling alone or in smaller families had comparatively lower fatalities.
- Family size influenced evacuation efficiency.

5. Classes and Deaths by Embarked Port

This clustered column chart compares deaths by embarkation location.

Embarked Port	Total Deaths
Southampton (S)	353
Cherbourg (C)	51
Queenstown (Q)	20

Insights

- Southampton recorded the highest number of deaths because it had the largest passenger population.
- Queenstown contributed the fewest fatalities.
- Passenger distribution across embarkation ports directly influenced death counts.

6. Male vs Female Deaths by Passenger Class

This line chart compares fatalities for males and females across all passenger classes.

Class	Male Deaths	Female Deaths
Third Class	215	55
Second Class	84	6
First Class	61	3

Insights

- Male fatalities exceeded female fatalities in every passenger class.
- Third Class experienced the highest mortality for both genders.
- First Class recorded the lowest fatalities, highlighting the survival advantage of higher-class passengers.

Major Findings

Gender Analysis

- Male passengers represented **approximately 85% of all deaths**.
- Female passengers experienced significantly fewer fatalities, supporting the historical **"Women and Children First"** evacuation strategy.

Age Group Analysis

- Adult passengers accounted for the highest number of deaths.
- Children experienced comparatively fewer fatalities due to rescue priority.
- Senior passengers recorded moderate fatalities because they represented a smaller portion of the passenger population.

Passenger Class Analysis

- Third Class passengers faced the highest mortality because of limited access to lifeboats and greater distance from evacuation points.
- First Class passengers benefited from better cabin locations and quicker access to rescue boats.

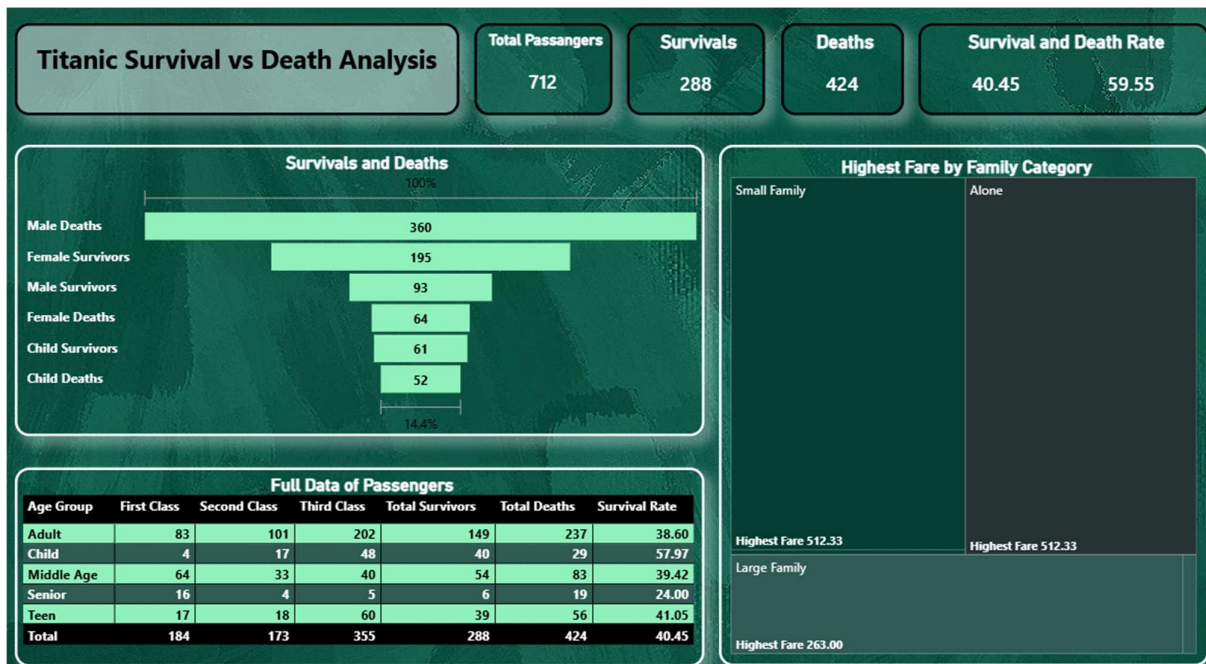
Family Category Analysis

- Large families encountered greater evacuation challenges, leading to higher child mortality.
- Smaller family groups generally had better evacuation outcomes.

Embarkation Analysis

- Southampton contributed the largest share of fatalities due to having the highest number of passengers boarding there.

Dashboard – 3



Dashboard – 3

Titanic Survival vs Death Analysis Dashboard

Overall Summary

- Total Passengers: 712
- Total Survivors: 288
- Total Deaths: 424
- Overall Survival Rate: 40.45%
- Overall Death Rate: 59.55%

Key Insights

1. Overall Survival vs Death

- Out of 712 passengers, 288 survived while 424 passengers died.
- The death rate (59.55%) is significantly higher than the survival rate (40.45%).

2. Gender-Based Analysis

- Male Deaths: 360
- Male Survivors: 93
- Female Survivors: 195
- Female Deaths: 64

Insight:

- Male passengers suffered the highest fatalities.
- Female passengers had a much higher survival count than males, reflecting the historical "Women and Children First" rescue policy.

3. Age Group Analysis

Age Group	Survivors	Deaths	Survival Rate
Adult	149	237	38.60%
Child	40	29	57.97%
Middle Age	54	83	39.42%
Teen	39	56	41.05%
Senior	6	19	24.00%

Insights:

- Adults had the highest number of both survivors and deaths because they formed the largest passenger group.
- Children achieved the highest survival rate (57.97%).
- Seniors had the lowest survival rate (24%).

4. Passenger Class Analysis

Passenger Class Survivors

First Class 184

Second Class 173

Third Class 355

Insight:

- Third Class had the largest number of passengers.
- Passenger class strongly influenced survival, with higher-class passengers generally having better access to lifeboats.

5. Highest Fare by Family Category

Family Category	Highest Fare
Small Family	512.33
Alone	512.33
Large Family	263.00

Insight:

- Passengers traveling alone or in small families paid the highest fares.
- Large families generally paid lower maximum fares.

Major Findings

1. The dashboard analyzes 712 Titanic passengers.
2. 288 passengers survived, while 424 passengers died.
3. The overall survival rate is 40.45%, whereas the death rate is 59.55%.
4. Male passengers experienced the highest number of deaths (360).
5. Female passengers recorded more survivors (195) than deaths (64).
6. Children had the highest survival rate (57.97%), indicating rescue priority.
7. Senior passengers had the lowest survival rate (24%).
8. Adults recorded the highest number of both survivors (149) and deaths (237) due to their larger population.
9. Passengers traveling alone or in small families paid the highest ticket fares (512.33).
10. Gender, age group, passenger class, family size, and ticket fare were the key factors influencing survival and death outcomes.

Conclusion

The Titanic Survival vs Death Analysis dashboard provides a complete comparison of survival and fatality patterns among Titanic passengers. The analysis shows that gender, age, passenger class, family size, and fare had a significant impact on survival. Female passengers and children had better survival outcomes, while adult males experienced the highest fatalities. Overall, the dashboard effectively summarizes the major factors that influenced survival during the Titanic disaster through clear KPIs, charts, and detailed data tables.

Report

REPORT

- A total of 712 passengers were analyzed in the Titanic dataset.
- 288 passengers survived, resulting in an overall survival rate of 40.45%.
- 424 passengers lost their lives, giving a death rate of 59.55%.
- Adult passengers recorded the highest number of survivors (149), making them the largest surviving age group.
- Children had the highest survival rate (57.97%), indicating they were given priority during evacuation.
- Senior passengers had the lowest survival rate (24.00%), making them the most vulnerable age group.
- Male passengers accounted for the highest number of deaths (360), while female deaths were only 64, showing a significant gender difference.
- Third Class passengers experienced the highest number of deaths (270), suggesting lower-class passengers faced greater evacuation challenges.
- Southampton (S) recorded the highest number of passenger deaths (353), as it had the largest number of passengers boarding.
- Passenger class, gender, age group, and family size were the major factors influencing survival and death outcomes during the Titanic disaster.